GROUP PROCESS DOCUMENTATION

Chillin-Digiphant

Digital twins, movement controls, and a shared racing game

Team: Alex Li, Stephen Jaja, and Ryanne Plaisance
Project: DigiPhant Kart / Rainbow Savannah

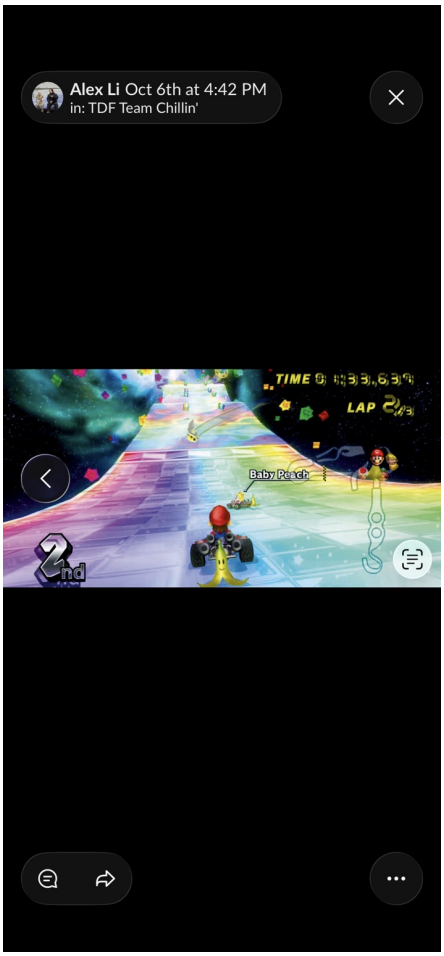


Team members trying movement gestures together. Players coordinate different inputs to control a single elephant kart.

This document covers the project from digital twin exploration through the obstacle course, game design, and group testing. It includes process evidence, control iterations, and two diagrams explaining how the game works. The current control mapping is based on the October 13 repository instructions; earlier experiments remain visible as part of the development process.

From digital twins to a racing game

Exploring digital twins introduced the idea of using real-world data to inform a virtual system. Examples such as aircraft testing and earthquake simulation showed how digital representations can support experimentation. Our elephant project used live movement data to drive an interactive Unity scene.



Rainbow Road reference shared by Alex on October 6. Inspiration image, not a screenshot of the team's game.

The team began with a supplied obstacle course, then considered games with similar play that we enjoyed. Mario Kart was a shared reference: its recognizable racing format fit the obstacle course brief. Rainbow Road inspired the direction.

An earlier savanna experience developed into Peel Patrol, where oranges were used to defend a watering hole. The project then shifted toward racing while preserving the earlier scenes.

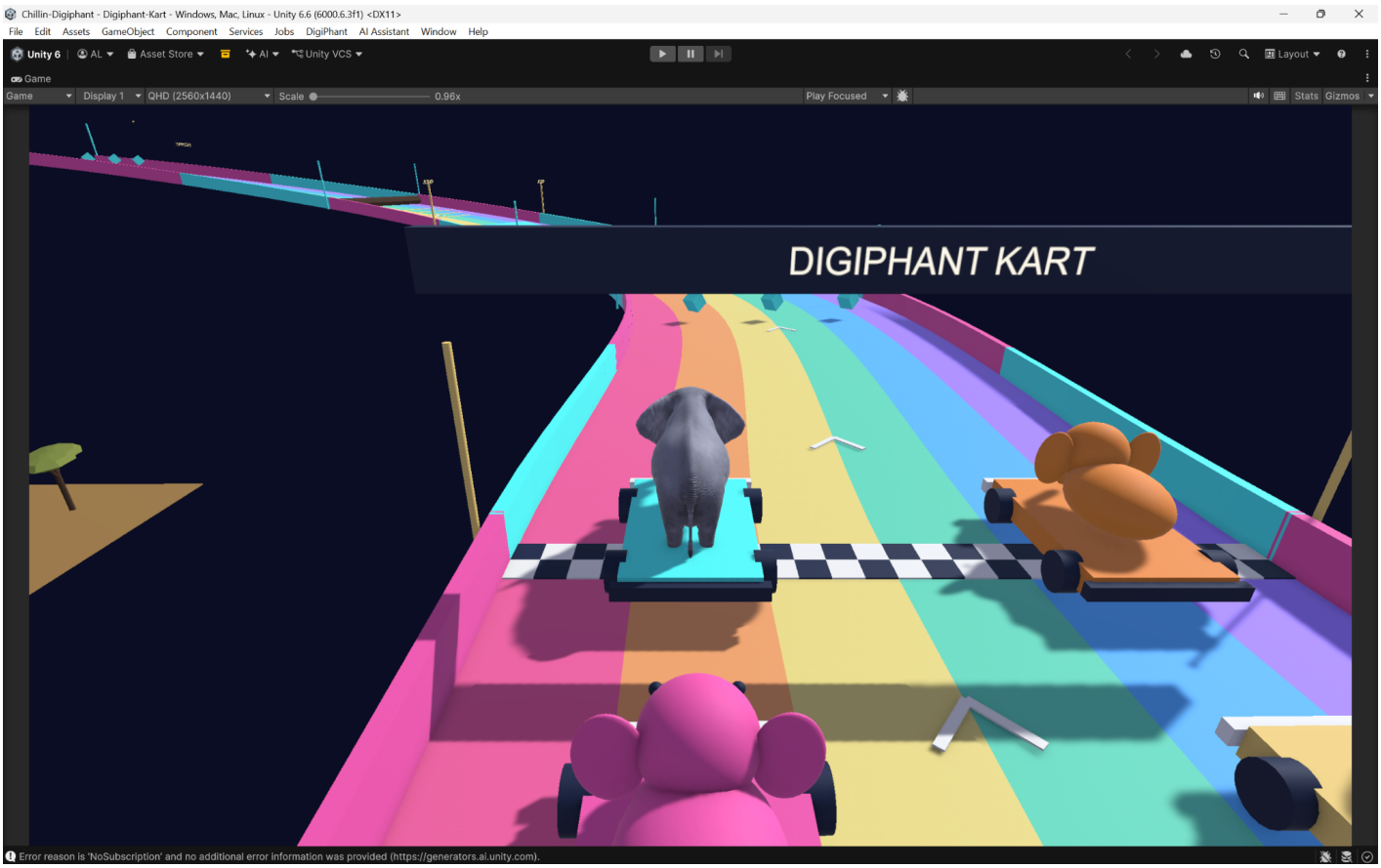
The racing game added pickups, rivals, checkpoints, collisions, and obstacles requiring players to jump or crouch. The oranges were discussed together. Ryanne suggested music as "the icing on the cake."

Design progression
Savanna play → Peel Patrol → elephant racing

The course became a reason for players to coordinate: movement, steering, and actions needed to work together rather than simply demonstrate isolated gestures.

Turning the reference into a scene

The Rainbow Road reference became a rainbow raceway in the team’s Unity scene. The screenshot shows the elephant kart, rival racers, directional markers, and a checkered starting line. These visible elements document the shift from the earlier savanna game to racing.



DigiPhant Kart in Unity. This is the team’s game scene, distinct from the Mario Kart reference shown on the previous page. It provides visual evidence of the implemented racing direction.

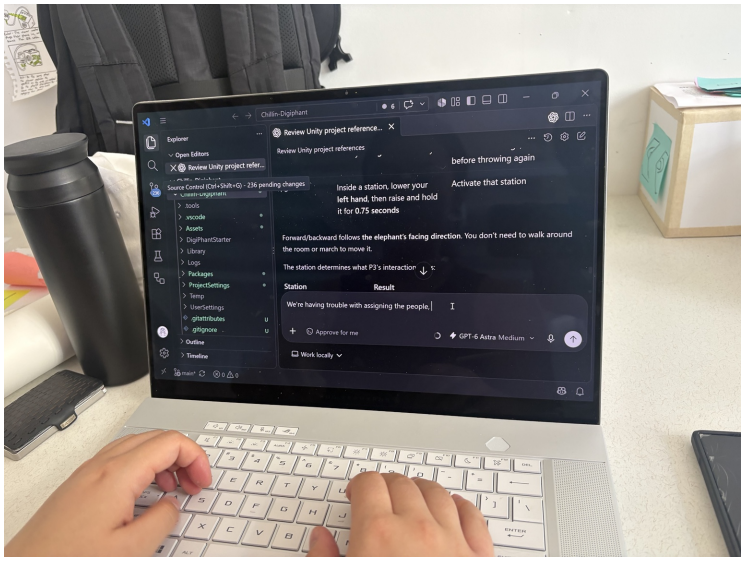
What changed: A watering-hole defense setting became a colorful raceway with multiple karts.
What stayed: The elephant remained central, and players continued using coordinated movements to control the experience.

Testing the controls together

The first major tracking problem was that the system detected only two people at once. The team worked through this by giving each person a separate box in the camera view. Ryanne reported that it then recognized all three and worked well during that testing.



Earlier Peel Patrol test. The interface shows the game, live camera preview, and three-performer mode.



Troubleshooting in VS Code.
The visible Codex prompt identifies a problem with assigning people to controls. The team used software changes and physical testing to work through it.

The key refinement was separating player input regions so the system could distinguish the performers.

Refining movement and feedback

1. Recognizing the team
Problem: only two performers were detected.
Change: divide the camera view into separate player regions.
Observed result: all three were recognized in the team's reported test.

2. Making jumping practical
Ryanne's earlier role included moving forward, backward, and jumping. Physically jumping was unreliable and tiring. The team changed her jump input to a leg lift. The current documented version moves jumping to P3's peace sign.

3. Giving the game a racing identity
Rainbow Road informed the design direction. Oranges added an interaction the team discussed together. Music was suggested to complete the experience. The current game uses an original synthesized score and action sounds.

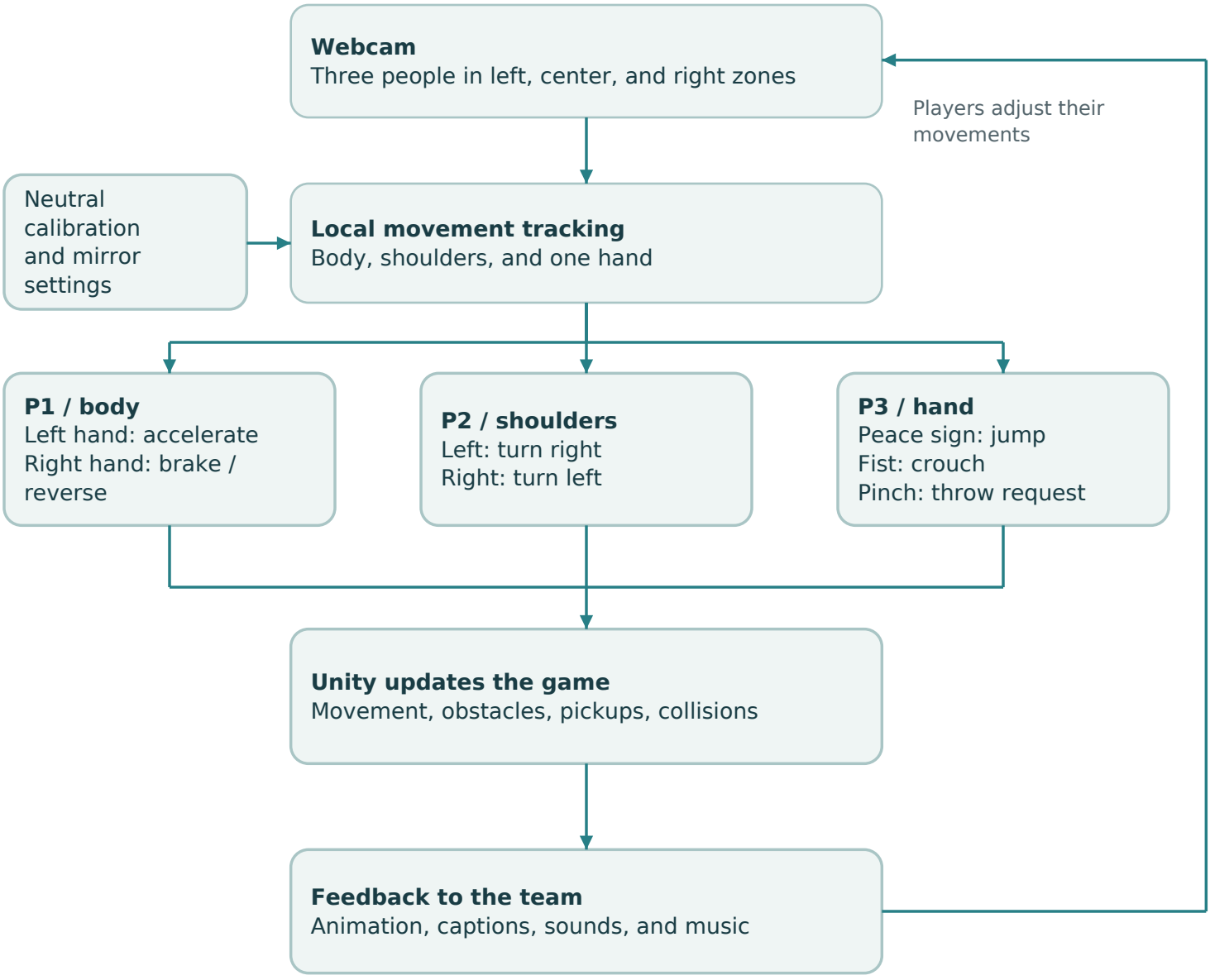


Coordination as part of play.
Each person controls a different part of the same kart. Ryanne described the teamwork as both fun and rewarding once tracking worked.

This photo shows group gesture practice. Its capture date has not been confirmed.

How three players control one kart

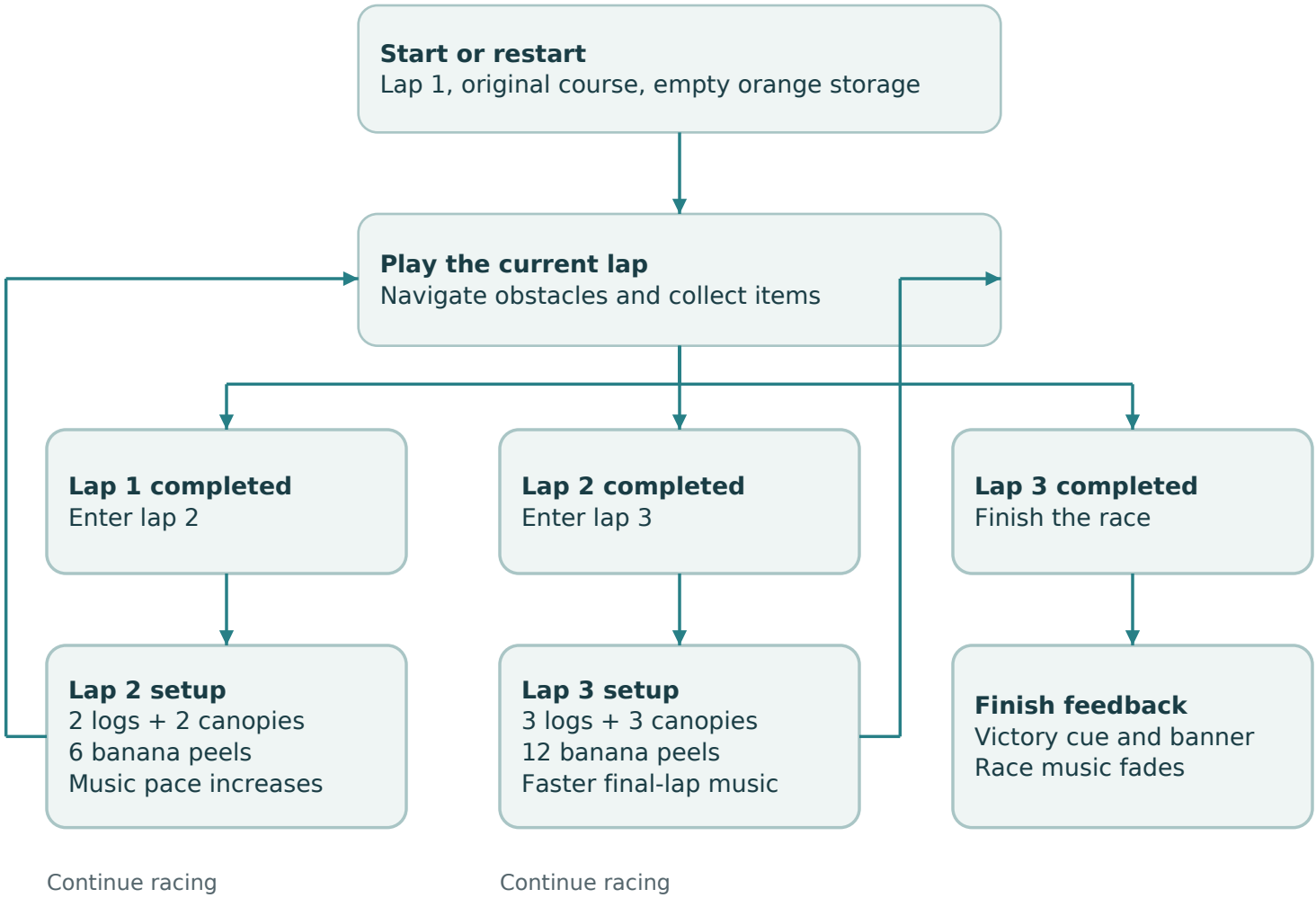
The current documented system uses a webcam and local tracking. Player regions and neutral calibration help translate separate movements into actions on one kart.



Conditions: Missing tracking can hold driving or disable an affected action. Throwing requires collected oranges. Fingers must relax between peace-sign jumps. Changed camera settings require new neutral calibration. This diagram shows the current version, not every earlier control experiment.

How the race changes with each lap

A completed lap changes both the next set of obstacles and the feedback players receive. The diagram shows the documented game rules, rather than the visual layout of the track.



Rules that apply during play: Each orange pickup adds one orange up to a capacity of three; each throw uses one. Banana contact triggers a brief slowdown and visual feedback. Restart returns the course to its first-lap setup. Newly added hazards wait when a racer is too close to their spawn location.

Source: Shared project README and current controls
Installation: Installation and troubleshooting guide
Diagrams prepared October 13, 2026. Technical rules describe the documented version; they do not establish that every feature was physically tested by every team member.